

Lecture 16

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Algorithms for constrained optimization

For simplicity, we will write an equality constrained optimization problem in the form

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ \text{subject to} & G(\mathbf{x}) = \mathbf{0}\end{array}$$

where $G : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the function $G(\mathbf{x}) = (g_1(\mathbf{x}), \dots, g_m(\mathbf{x}))^\top$.

We will write an inequality constrained optimization problem in the form

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ \text{subject to} & H(\mathbf{x}) \leq \mathbf{0}\end{array}$$

where $H : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is the function $H(\mathbf{x}) = (h_1(\mathbf{x}), \dots, h_p(\mathbf{x}))^\top$ and where we have assumed that equality constraints, if any, have been expressed as equivalent inequality constraints.

The most widely used strategy for constrained optimization is to convert the problem to an unconstrained optimization problem and then use one of the algorithms for the latter:

- steepest descent
- quasi-Newton
- Newton

More specifically, we find an F or a sequence of F_k 's depending on f and G (or H) such that either:

- the unconstrained minimizer of F equals the desired constrained minimizer or
- the limit of the sequence unconstrained minimizers of F_k converges to the desired constrained minimizer.

Penalty function method for equality constrained optimization

We will start with the method of penalty function. Note that this depend on the optimization problem, i.e., on f and G .

The basic idea is to add a penalty term to the objective function that penalizes violations of the constraints.

We'll include a parameter ρ that controls the strength of the penalty.

What will success look like?

We'll need to find a point where small changes in $\mathbf{x}$ do not change the value of the objective function.

Moreover, we will need for the penalty term to be *strong enough* that the algorithm can't just decrease the value of the objective function by moving away from the constraint.

The portion of the gradient that is due to the penalty term will need to dominate the portion that is due to $f(\mathbf{x})$.

If our penalty is set up such that it's quite flat near the constraint, we'll need to multiply it by a big number ρ to get it to dominate!

In our first scenario, we will need to take the limit as $\rho \rightarrow \infty$ to find solutions that satisfy the constraints exactly.

The quadratic penalty function or l^2 -penalty function is

$$P_2(\mathbf{x}, \rho) = f(\mathbf{x}) + \frac{\rho}{2} G(\mathbf{x})^\top G(\mathbf{x}) = f(\mathbf{x}) + \frac{\rho}{2} \|G(\mathbf{x})\|_2^2 = f(\mathbf{x}) + \frac{\rho}{2} \sum_{i=1}^m g_i(\mathbf{x})^2$$

where $\rho \geq 0$.

As $\rho \rightarrow \infty$, the penalty term will blow up – unless $G(\mathbf{x}) = \mathbf{0}$.

So any minimization algorithm would avoid this by making sure that $\mathbf{x}$ is such that $g_i(\mathbf{x}) = 0$ for all $i = 1, \dots, m$, i.e., $G(\mathbf{x}) = \mathbf{0}$. This can be formalized and rigorously established.

Under reasonable assumptions, if we find the best $\mathbf{x}$ for each ρ , and then we take the limit as $\rho \rightarrow \infty$, we will get the best $\mathbf{x}$ that satisfies the constraints.

Formally,

Theorem

Let

$$\mathbf{x}_*(\rho) = \operatorname{argmin} P_2(\mathbf{x}, \rho) \quad \text{and} \quad \mathbf{x}_* = \operatorname{argmin}\{f(\mathbf{x}) : G(\mathbf{x}) = \mathbf{0}\}.$$

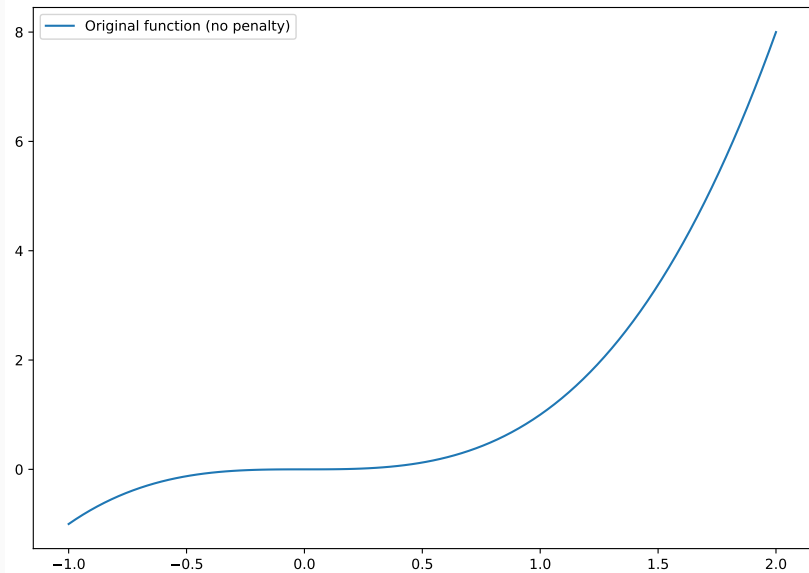
Under reasonably general conditions,

$$\lim_{\rho \rightarrow +\infty} \mathbf{x}_*(\rho) = \mathbf{x}_*$$

Example

The univariate optimization problem

$$\begin{array}{ll}\text{minimize} & x^3 \\ \text{subject to} & x - 1 = 0\end{array}$$



$$\begin{array}{ll}\text{minimize} & x^3 \\ \text{subject to} & x - 1 = 0\end{array}$$

What is the minimizer x_* of this problem? We can just do it in our head:

The minimizer is $x_* = 1$. That's the only point that satisfies the constraint.

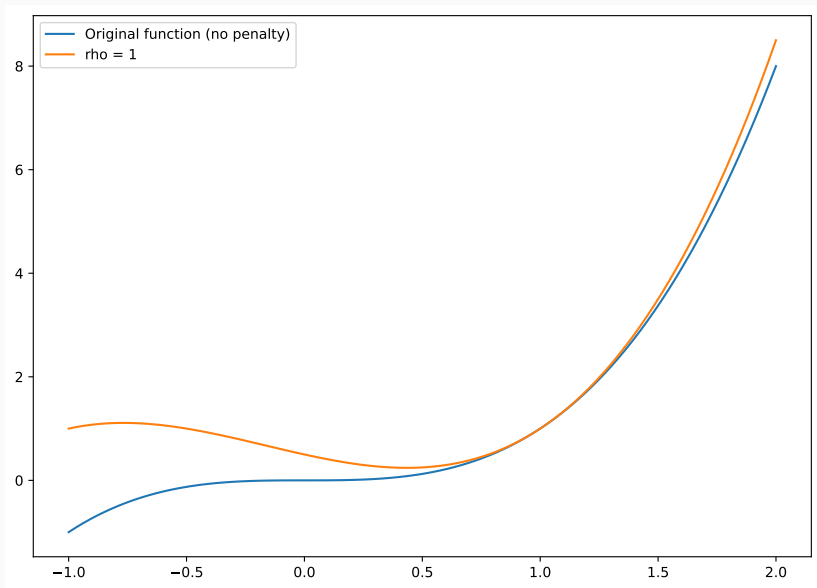
$$\begin{array}{ll}\text{minimize} & x^3 \\ \text{subject to} & x - 1 = 0\end{array}$$

But let's try to solve it using the penalty function method.

We write down the quadratic penalty function:

$$P_2(x, \rho) = x^3 + \frac{\rho}{2}(x - 1)^2$$

Let's plot this function for $\rho = 1$.



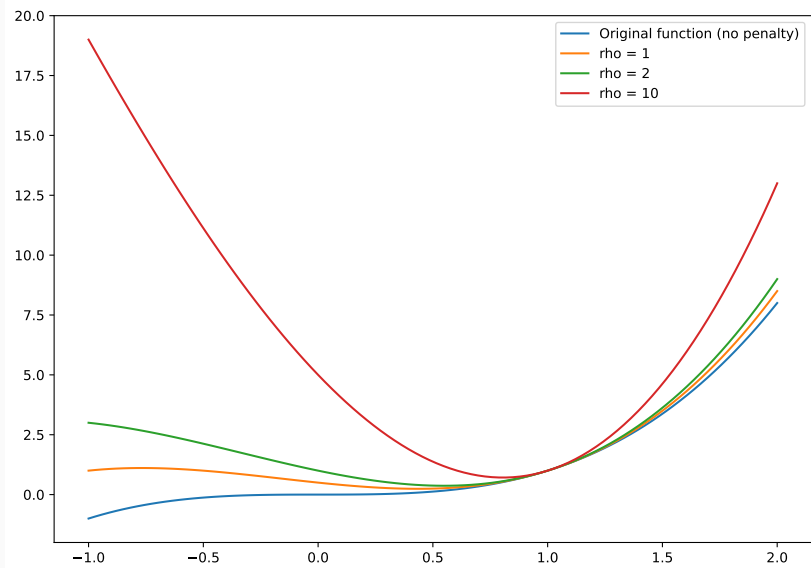
We see that the penalty function increased the value of our objective function away from the constraint, so that the constraint starts to become a minimum.

However, it didn't do very much right at the constraint (the two curves are almost identical). Why?

$$P_2(x, \rho) = x^3 + \frac{\rho}{2}(x - 1)^2$$

The quadratic term is very small when x is close to 1!

Let's look for a few more values of ρ :

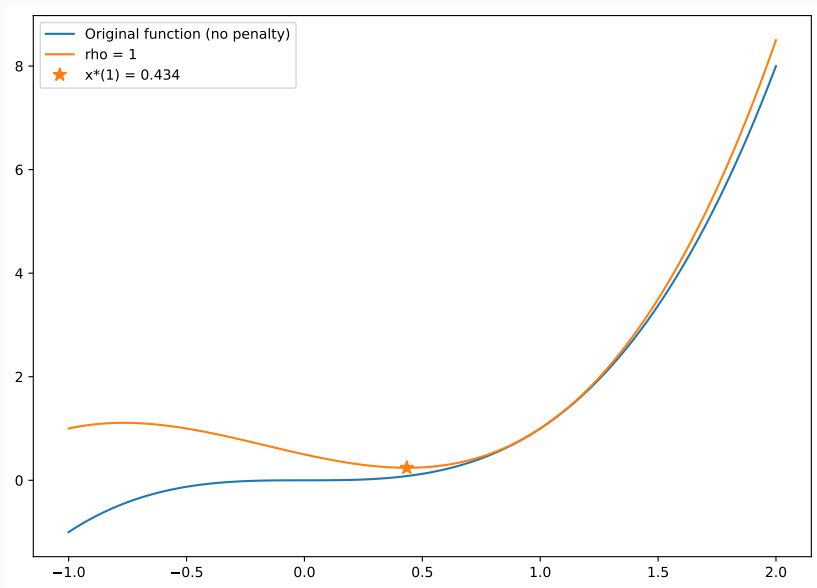


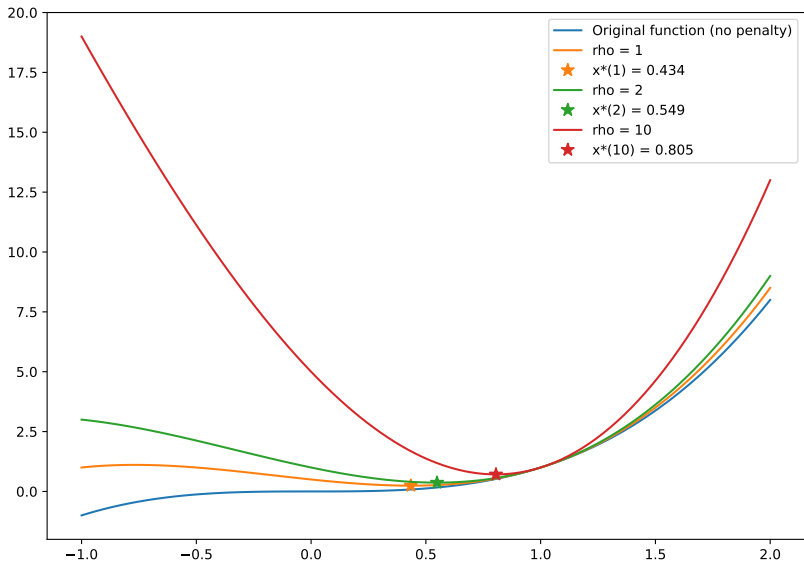
As ρ increases, the penalty function starts to have an effect even close to the constraint.

OK, let's remember our procedure. We are going to find the unconstrained minimizer $x_*(\rho)$ for each $\rho \geq 0$ and then take the limit as $\rho \rightarrow \infty$.

Now, holding $\rho \geq 0$ constant for now, we want to find the *unconstrained* minimizer $x_*(\rho)$.

Let's look at the graphs again, this time with the minimizers marked:





The minimizers are getting closer and closer to the constraint as ρ increases! Great!

Now let's do it analytically.

Remember that our objective function is

$$P_2(x, \rho) = x^3 + \frac{\rho}{2}(x - 1)^2$$

First, we find stationary points. We differentiate with respect to x and set the derivative equal to zero:

$$\frac{d}{dx} P_2(x, \rho) = 3x^2 + \rho(x - 1) = 0$$

Using the quadratic equation, there are two possible solutions:

$$x_*(\rho) = \frac{-\rho + \sqrt{\rho^2 + 12\rho}}{6} \quad \text{and} \quad x_*(\rho) = \frac{-\rho - \sqrt{\rho^2 + 12\rho}}{6}$$

$$x_1(\rho) = \frac{-\rho + \sqrt{\rho^2 + 12\rho}}{6} \quad \text{and} \quad x_2(\rho) = \frac{-\rho - \sqrt{\rho^2 + 12\rho}}{6}$$

Which one is a minimizer? Well, we can take the second derivative to see:

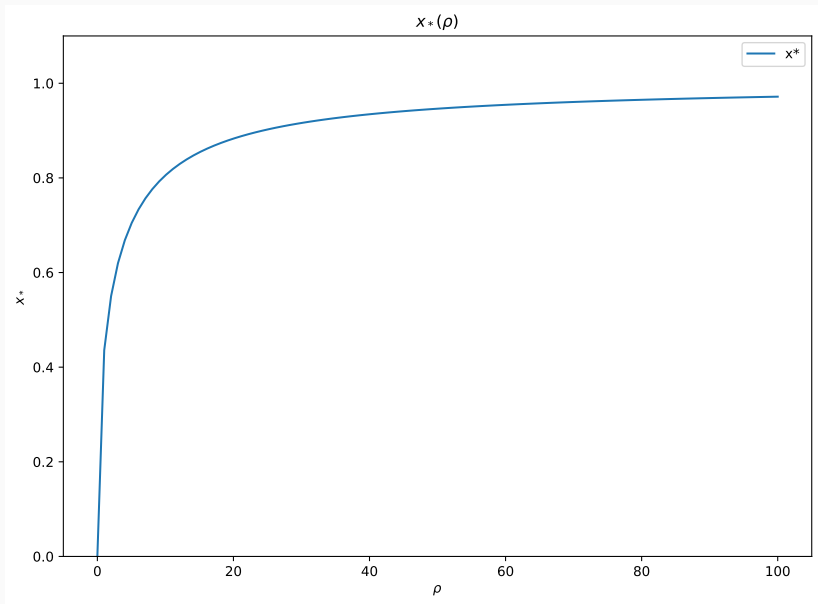
$$\frac{d^2}{dx^2} P_2(x, \rho) = 6x + \rho$$

Finding the second derivative at each of the two solutions, we get:

$$\frac{d^2}{dx^2} P_2(x_1(\rho), \rho) = 6x_1(\rho) + \rho = 6 \left(\frac{-\rho + \sqrt{\rho^2 + 12\rho}}{6} \right) + \rho = \sqrt{\rho^2 + 12\rho} > 0$$

$$\frac{d^2}{dx^2} P_2(x_2(\rho), \rho) = 6x_2(\rho) + \rho = 6 \left(\frac{-\rho - \sqrt{\rho^2 + 12\rho}}{6} \right) + \rho = -\sqrt{\rho^2 + 12\rho} < 0$$

So $x_1(\rho)$ is a local minimizer. Therefore we have found $x_*(\rho) = x_1(\rho) = \frac{-\rho + \sqrt{\rho^2 + 12\rho}}{6}$



But remember our initial problem was this:

$$\begin{array}{ll}\text{minimize} & x^3 \\ \text{subject to} & x - 1 = 0\end{array}$$

To find the minimizer that actually satisfies the constraint, we need to take the limit as $\rho \rightarrow \infty$:

$$\lim_{\rho \rightarrow \infty} \frac{-\rho + \sqrt{\rho^2 + 12\rho}}{6} = \lim_{\rho \rightarrow \infty} \frac{\rho(-1 + \sqrt{1 + \frac{12}{\rho}})}{6}$$

We know from the graph that this limit is 1, but let's do it more formally.

$$\lim_{\rho \rightarrow \infty} \frac{-\rho + \sqrt{\rho^2 + 12\rho}}{6} = \lim_{\rho \rightarrow \infty} \frac{\rho(-1 + \sqrt{1 + \frac{12}{\rho}})}{6}$$

Using the Taylor expansion of $\sqrt{1 + \frac{12}{\rho}}$ around $\frac{12}{\rho} = 0$:

$$\sqrt{1 + \frac{12}{\rho}} \approx 1 + \frac{6}{\rho} + O\left(\frac{1}{\rho^2}\right)$$

The higher order terms go to zero as $\rho \rightarrow \infty$.

Substituting back:

$$x_* = \lim_{\rho \rightarrow \infty} x_*(\rho) = \lim_{\rho \rightarrow \infty} \frac{\rho(-1 + 1 + \frac{6}{\rho})}{6} = \lim_{\rho \rightarrow \infty} \frac{\rho(\frac{6}{\rho})}{6} = 1$$

Good! After all that work, we got the same answer using the penalty function method as we did by solving the problem directly.

The quadratic penalty function does a good job of keeping us away from values that strongly violate the constraints – that's what the square does.

However, when we are close but not equal to the constraint, the quadratic penalty flattens out. It doesn't penalize us heavy for small violations.

This is why we need to take the limit as $\rho \rightarrow \infty$ to get the true minimizer.

A different approach is to use a penalty function that doesn't flatten out close to the constraint – for example, the absolute value function.

We will no longer need to take the limit as $\rho \rightarrow \infty$ to get the true minimizer. However, we lose differentiability!

The absolute value penalty function or l^1 -penalty function is defined by

$$P_1(\mathbf{x}, \rho) = f(\mathbf{x}) + \rho \|G(\mathbf{x})\|_1 = f(\mathbf{x}) + \rho \sum_{i=1}^m |g_i(\mathbf{x})|$$

ρ does not need to be arbitrarily large for $\mathbf{x}_*(\rho)$ to become $\mathbf{x}_*$.

However it suffers from the problem of not being differentiable in general - requiring us to use nondifferentiable optimization algorithms.

Roughly speaking, these are algorithms that use subgradients in place of the gradient.

(Remember, a **subgradient** is a generalization of the gradient for nondifferentiable functions; it's a straight line or hyperplane that touches the function at our point and lies below the function in a neighborhood of our point.)

Theorem

Let

$$\mathbf{x}_*(\rho) = \operatorname{argmin} P_1(\mathbf{x}, \rho) \quad \text{and} \quad \mathbf{x}_* = \operatorname{argmin}\{f(\mathbf{x}) : G(\mathbf{x}) = \mathbf{0}\}.$$

Under mild conditions, there exists $\bar{\rho} > 0$ such that

$$\mathbf{x}_*(\rho) = \mathbf{x}_*$$

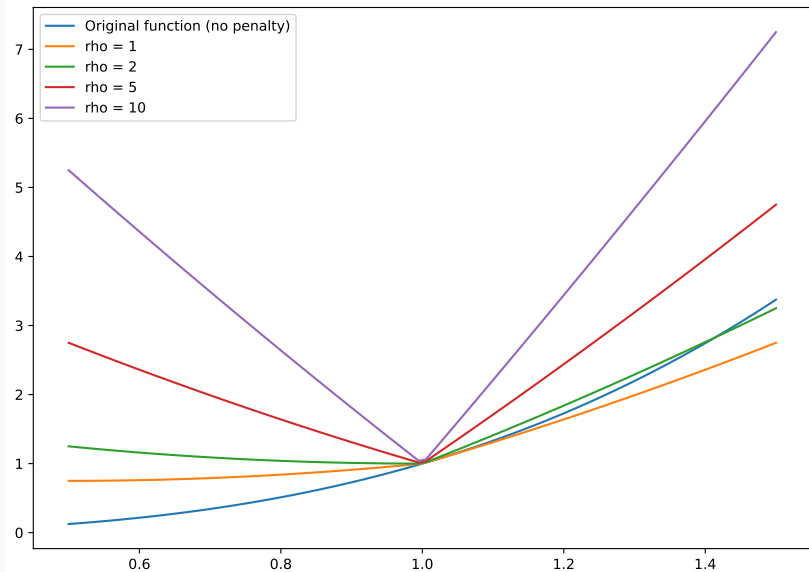
whenever $\rho > \bar{\rho}$.

Example

The univariate optimization problem

$$\begin{array}{ll}\text{minimize} & x^2 \\ \text{subject to} & x - 1 = 0\end{array}$$

has obvious minimizer $x_* = 1$



The l^1 -penalty function

$$P_1(x, \rho) = x^2 + \rho|x - 1|$$

has unconstrained minimizer $x_*(\rho) = 1$ if $\rho > 2 =: \bar{\rho}$.

For an equality constrained optimization problem, the Lagrangian takes the form

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda^\top G(\mathbf{x}).$$

As we saw in the last lecture, a necessary condition for minimizing $f(\mathbf{x})$ subject to $G(\mathbf{x}) = 0$ is the stationarity condition in KKT: $\nabla L(\mathbf{x}_*, \lambda_*) = \mathbf{0}$.

How can we find points where this is satisfied?

We have a lot of techniques for finding local minima of functions like $L(\mathbf{x}, \lambda)$. And we know that at a local minimum, $\nabla L(\mathbf{x}_*, \lambda_*) = \mathbf{0}$.

Unfortunately, the converse is not true. Our Lagrangian might not have a minimum at $(\mathbf{x}_*, \lambda_*)$... it may be at a saddle point or maximizer instead. (We know that f will be increasing as we move away from our point, but G might actually be decreasing!)

We will need a slightly different approach, something that takes into account the magnitude of $G(\mathbf{x})$, not just its actual value. But we want to preserve the ability to check for stationarity.

Formally, when $\lambda = \lambda_*$, the critical Lagrange multiplier, the stationarity condition in KKT conditions says that

$$\nabla L(\mathbf{x}_*, \lambda_*) = \mathbf{0},$$

i.e., $\mathbf{x}_*$ is a stationary point of $L(\mathbf{x}, \lambda_*)$. However,

$$\nabla^2 L(\mathbf{x}_*, \lambda_*) = \nabla^2 f(\mathbf{x}_*) + \sum_{i=1}^m \lambda_i^* \nabla^2 g_i(\mathbf{x}_*)$$

may not be positive semidefinite.

So $\mathbf{x}_*$ is generally not a local minimizer of $L(\mathbf{x}, \lambda_*)$.

However, if we add a quadratic penalty term to $L(\mathbf{x}, \lambda)$, we obtain the augmented Lagrangian

$$L_A(\mathbf{x}, \lambda, \rho) = f(\mathbf{x}) + \lambda^\top G(\mathbf{x}) + \frac{\rho}{2} G(\mathbf{x})^\top G(\mathbf{x})$$

where $\rho \geq 0$.

The quadratic penalty term preserves the stationarity of $\mathbf{x}_*$ as

$$\nabla L_A(\mathbf{x}_*, *, \rho) = \nabla L(\mathbf{x}_*, *) + \rho JG(\mathbf{x}_*)^\top G(\mathbf{x}_*) = 0$$

by the stationarity ($\nabla L(\mathbf{x}_*, *) = \mathbf{0}$) and primal feasibility ($G(\mathbf{x}_*) = \mathbf{0}$) KKT conditions, i.e., $(\mathbf{x}_*, *, \rho)$ is a stationary point of L_A for any $\rho \geq 0$.

As

$$\nabla^2 L_A(\mathbf{x}, \lambda, \rho) = \nabla^2 f(\mathbf{x}) + \sum_{i=1}^m (\lambda_i + \rho g_i(\mathbf{x})) \nabla^2 g_i(\mathbf{x}) + \rho \mathbf{J}G(\mathbf{x})^\top \mathbf{J}G(\mathbf{x})$$

and $g_i(\mathbf{x}_*) = 0$ for all $i = 1, \dots, m$, we get

$$\nabla^2 L_A(\mathbf{x}_*, \lambda^*, \rho) = \nabla^2 f(\mathbf{x}_*) + \sum_{i=1}^m \lambda_i^* \nabla^2 g_i(\mathbf{x}_*) + \rho \mathbf{J}G(\mathbf{x}_*)^\top \mathbf{J}G(\mathbf{x}_*).$$

Note that $JG(\mathbf{x}_*)^\top JG(\mathbf{x}_*)$ is a positive semidefinite matrix. Under certain conditions, when ρ is large enough, $\nabla^2 L_A(\mathbf{x}_*, *, \rho)$ can be guaranteed to be positive definite and thus $\mathbf{x}_*$ becomes a minimizer of $L_A(\mathbf{x}, *, \rho)$.

Theorem

Let

$$(\mathbf{x}_*(\rho), *(\rho)) = \operatorname{argmin} L_A(\mathbf{x}, *, \rho) \quad \text{and} \quad \mathbf{x}_* = \operatorname{argmin}\{f(\mathbf{x}) : G(\mathbf{x}) = \mathbf{0}\}.$$

Under mild conditions, there exists $\bar{\rho} > 0$ such that

$$\mathbf{x}_*(\rho) = \mathbf{x}_*$$

whenever $\rho > \bar{\rho}$.

Example

The univariate optimization problem

$$\begin{aligned} &\text{minimize } x^3 \\ &\text{subject to } x + 1 = 0 \end{aligned}$$

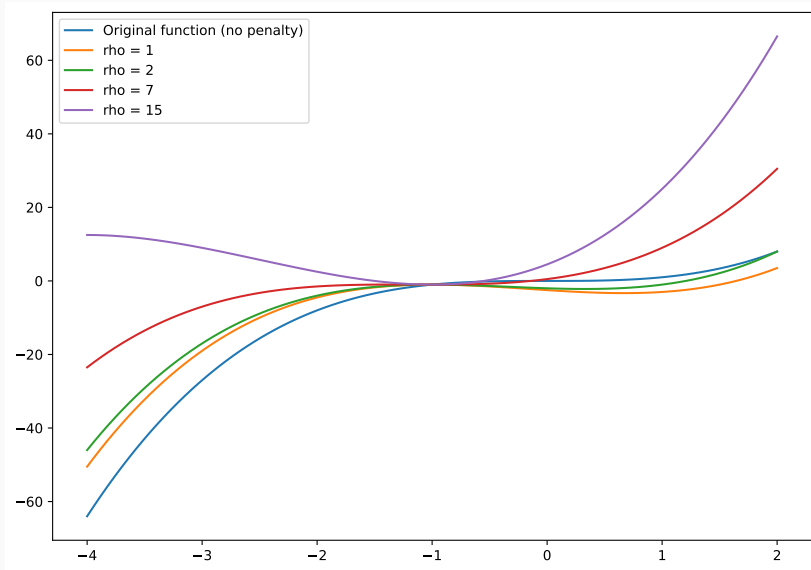
has obvious minimizer $x_* = -1$ and critical Lagrange multiplier $\lambda_* = -3$. The Lagrangian

$$L(x, \lambda_*) = x^3 - 3(x + 1)$$

has $x_* = -1$ as an unconstrained maximizer (not minimizer!). However the augmented Lagrangian

$$L_A(x, \lambda_*, \rho) = x^3 - 3(x + 1) + \frac{\rho}{2}(x + 1)^2$$

has local minimizer $x_* = -1$ if $\rho > 6 =: \bar{\rho}$.



For any of these methods that involve a penalty parameter ρ , the way you implement them is to pick an increasing sequence of parameters $\rho_1 < \rho_2 < \rho_3 < \dots$, e.g., $\rho_k = 2^k$ or $\rho_k = 10^k$. We then use the k th minimizer $\mathbf{x}_*(\rho_k)$ as the starting point of the $(k+1)$ th iteration, i.e.

$$\mathbf{x}_0(\rho_{k+1}) = \mathbf{x}_*(\rho_k).$$

If we just pick an excessively large value of ρ or a sequence of ρ_k 's that increases too fast, the method will fail. Choosing an appropriate sequence is an empirical process that depends on many factors - more an art that comes from experience than a science.

Inequality constrained optimization

For the inequality constrained optimization problem

$$\begin{array}{ll} \text{minimize } f(\mathbf{x}) & \text{or} \quad \text{minimize } f(\mathbf{x}) \\ \text{subject to } H(\mathbf{x}) \leq \mathbf{0}, & \text{subject to } h_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, p, \end{array}$$

we introduce the following terminologies:

Definition

With respect to a constraint $h_i(\mathbf{x}) \leq 0$ and a point $\mathbf{x} \in \mathbb{R}^n$, the point $\mathbf{x}$ is said to be

- feasible if $h_i(\mathbf{x}) \leq 0$,
 - infeasible if $h_i(\mathbf{x}) > 0$;
- and the constraint $h_i(\mathbf{x}) \leq 0$ is said to be
- active if $h_i(\mathbf{x}) = 0$,
 - inactive if $h_i(\mathbf{x}) < 0$,
 - violated if $h_i(\mathbf{x}) > 0$.

The feasible region is the set $\{\mathbf{x} : H(\mathbf{x}) \leq \mathbf{0}\}$ and its complement is called the infeasible region.

As we saw earlier that every equality constrained optimization problem can be formulated as an inequality constrained optimization problem by introducing more constraints.

It turns out that every inequality constrained optimization problem can also be formulated as an equality constrained optimization problem by introducing more variables. By simply observing that

$$h_i(\mathbf{x}) \leq 0 \quad \Leftrightarrow \quad h_i(\mathbf{x}) + s_i = 0, s_i \geq 0.$$

The variable s_i is called a **slack variable**.

By introducing one slack variable for each inequality constraint $s_1, \dots, s_p$, we may write an inequality constrained optimization problem as an equality constrained optimization problem with nonnegative variables:

$$\begin{aligned} & \text{minimize } f(\mathbf{x}) \\ & \text{subject to } h_i(\mathbf{x}) + s_i = 0, \quad i = 1, \dots, p, \\ & \quad \mathbf{s} \geq \mathbf{0} \end{aligned}$$

This is nearly as good as an equality constrained optimization problem, which means all previously discussed methods for equality constrained optimization apply verbatim. But note that while there were n optimization variables $x_1, \dots, x_n$ in the original problem, there are now $n + p$ optimization variables $x_1, \dots, x_n, s_1, \dots, s_p$ in the new problem.

However, if one insists on a purely equality constrained optimization problem, then we may write our slack variable in the form s_i^2 , noting that this is always nonnegative and so we do not need to impose nonnegativity constraint, giving us

$$\begin{aligned} & \text{minimize } f(\mathbf{x}) \\ & \text{subject to } h_i(\mathbf{x}) + s_i^2 = 0, \quad i = 1, \dots, p. \end{aligned}$$

Nevertheless the form in (16.8) is more common and preferred. For example, a common use of slack variables is to transform linear inequality constraints in linear programming from one form to another:

$$A\mathbf{x} \leq \mathbf{b} \quad \longrightarrow \quad \begin{cases} A\mathbf{x} + \mathbf{s} = \mathbf{b} \\ \mathbf{s} \geq \mathbf{0} \end{cases}$$

In order to get a linear programming problem after the transformation, we have to keep all variables linear so we cannot introduce quadratic terms like s_i^2 .

Let $\text{ReLU}(t) := \max(t, 0)$ and extend it to vector variables in a coordinatewise sense:
for any $\mathbf{x} \in \mathbb{R}^n$,

$$\text{ReLU}(\mathbf{x}) := \begin{bmatrix} \text{ReLU}(x_1) \\ \vdots \\ \text{ReLU}(x_n) \end{bmatrix}.$$

Observe that $\text{ReLU}(H(\mathbf{x}))$ keeps track of the violated constraints. With this, it is straightforward to modify the penalty function method for inequality constraints - just replace G with $\text{ReLU}(H)$. The quadratic and l^1 -penalty functions are given by

$$P_2(\mathbf{x}, \rho) = f(\mathbf{x}) + \frac{\rho}{2} \|\text{ReLU}(H(\mathbf{x}))\|_2^2, \quad P_1(\mathbf{x}, \rho) = f(\mathbf{x}) + \rho \|\text{ReLU}(H(\mathbf{x}))\|_1.$$

Note that minimizing these for sufficiently large ρ have the effect of forcing $\mathbf{x}$ to move into the feasible region so that there are no violated constraint and the penalty term vanishes.

We may also state analogues of the respective theorems for inequality constrained optimization. In addition to

$$\lim_{\rho \rightarrow +\infty} \mathbf{x}_*(\rho) = \mathbf{x}_*,$$

one may also show

$$\lim_{\rho \rightarrow +\infty} \rho h_i(\mathbf{x}_*(\rho)) = -\lambda_i^*.$$

In particular it follows that $\lambda_i^* \geq 0$, i.e., the dual feasibility condition is automatically satisfied.

Unlike equality constrained optimization, we have to keep track of which constraints are violated and which are not. Ideally we want a method that avoids this. This consideration leads us to our next method.

While penalty function methods work by coercing $\mathbf{x}$ to move from infeasible region into feasible region, barrier function methods work by trapping $\mathbf{x}$ within the feasible region so that it never move into infeasible region.

Because of this, barrier function method requires that we start from an initial point $\mathbf{x}_0$ that is feasible, i.e., lies within the feasible region.

The two most common types of barrier functions are the log barrier:

$$b(\mathbf{x}, \mu) = f(\mathbf{x}) - \mu \sum_{i=1}^p \log(-h_i(\mathbf{x}))$$

and the inverse barrier:

$$b(\mathbf{x}, \mu) = f(\mathbf{x}) - \mu \sum_{i=1}^p \frac{1}{h_i(\mathbf{x})},$$

where $\mu \geq 0$.

Note that whenever $\mathbf{x}$ moves close to the boundary, i.e., where $h_i(\mathbf{x}) = 0$ for some $i \in \{1, \dots, p\}$, the barrier term blows up.

This produces the effect of never allowing $\mathbf{x}$ to escape into the infeasible region but the trouble is that if the minimizer $\mathbf{x}_*$ is near the boundary, $\mathbf{x}$ would kept away from it. The trick is to gradually decrease μ to zero.

Theorem

Let

$$\mathbf{x}_*(\mu) = \operatorname{argmin} b(\mathbf{x}, \mu) \quad \text{and} \quad \mathbf{x}_* = \operatorname{argmin}\{f(\mathbf{x}) : H(\mathbf{x}) \leq \mathbf{0}\}.$$

Under mild conditions,

$$\lim_{\mu \rightarrow 0^+} \mathbf{x}_*(\mu) = \mathbf{x}_*.$$

In fact, we may also show

$$\lim_{\mu \rightarrow 0^+} \frac{\mu}{h_i(\mathbf{x}_*(\mu))} = -\lambda_i^*.$$

Note that in particular it follows (why?) that $\lambda_i^* \geq 0$, i.e., the dual feasibility condition is automatically satisfied.

Example

The univariate optimization problem

$$\begin{aligned} &\text{minimize } x^2 \\ &\text{subject to } 1 - x \leq 0, \end{aligned}$$

has obvious minimizer $x_* = 1$ (exercise). The log barrier function

$$b(x, \mu) = x^2 - \mu \log(x - 1)$$

has unconstrained minimizer (exercise)

$$x_*(\mu) = \frac{1}{2} + \frac{1}{2}\sqrt{1 + 2\mu}$$

and clearly $x_*(\mu) \rightarrow 1$ as $\mu \rightarrow 0^+$.

Example

Consider the linearly constrained quadratic programming (LCQP) problem

$$\begin{aligned} & \text{minimize } x^2 - 2xy + 4y^2 \\ & \text{subject to } 1 - x/10 + y \leq 0 \end{aligned}$$

The log barrier function is

$$b(x, y, \mu) = x^2 - 2xy + 4y^2 - \mu \log(x/10 - y - 1).$$

Figure 18 shows the contour plots of the barrier function as μ decreases from 100 to 0.01 . Note that:

- the contours of $b(x, y, \mu)$ converge to those $f(x, y) := x^2 - 2xy + 4y^2$ as $\mu \rightarrow 0^+$;
- the feasible region is everything below the red line $h(x, y) := 1 - x/10 + y = 0$;
- all contours of $b(x, y, \mu)$ are confined to the feasible region.

Barrier functions are extremely important in convex optimization under the name interior point method, which is essentially Newton method applied to the barrier function. One of the biggest breakthrough in convex optimization in the 20th Century was the demonstration that if the feasible set has a barrier that satisfies a property called self-concordance, then the problem can be solved using Newton method to any ε -accuracy in polynomial time. Selfconcordant barriers exist for LP, convex QP, SOCP, SDP, GP and that's why they may all be solved efficiently. Examples of such barriers include

$$-\log(b - a_1x_1 - a_2x_2 - \cdots - a_nx_n), \quad a_1, \dots, a_n, b \in \mathbb{R}$$

for linear programming (LP) and

$$-\log \det(B - x_1A_1 - x_2A_2 - \cdots - x_nA_n), \quad A_1, \dots, A_n, B \in \mathbb{S}^m$$

for semidefinite programming (SDP). The latter is the reason we spent so much effort learning how to differentiate the $\log \det$ function.

For example, Homework 4, Problem 3 comes from solving an LP problem with $n = 100$ variables and $m = 500$ linear constraints:

$$\text{minimize } \mathbf{c}^\top \mathbf{x}$$